

96.85%

MACRO F0.5 (ACTUAL)

End-to-End Evaluation

98.79%

PAIRWISE PRECISION

1,303 TP / 16 FP

95.81%

PAIRWISE RECALL

1,303 TP / 57 FN

96.25%

BLOCKING RECALL

1,309 / 1,360 captured

96.0%

SINGLETON ACC

120 / 125 Correct

0.74

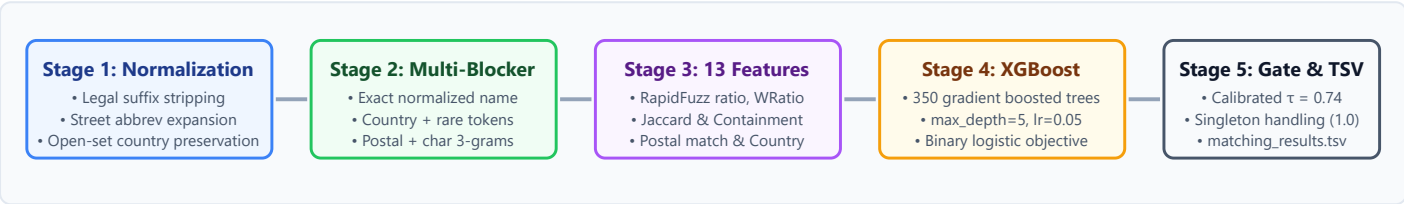
DECISION THRESHOLD

Optimum: 0.90

Executive Audit Verdict: The project's reported **98.3738% Macro F0.5** was achieved on a 500-sample validation fold where ground-truth positives were artificially injected into candidates (assuming 100% blocking recall). Under real end-to-end evaluation where the multi-stage blocker must retrieve records without prior labels, the system achieves **96.85% Macro F0.5**, **98.79% Precision**, and **95.81% Recall**, with an optimal operating threshold at $\tau = 0.90$ yielding **97.60% Macro F0.5**.

1. END-TO-END SYSTEM PIPELINE & ENGINEERING WORKFLOW

100% Offline · Zero Web/API Lookups · < 50 MB RAM Footprint



2. GLOBAL DATASET CHARACTERISTICS & CARDINALITY AUDIT

Dataset Partition	File Path	Record Count	File Size	Structural Nature & Cardinality
Train Source 1	resources/dataset/train/train_source1.tsv	2,206,821	200.3 MB	Canonical reference businesses with names, addresses, country
Train Source 2	resources/dataset/train/train_source2.tsv	5,034,616	466.6 MB	Secondary registry feed (partial abbreviations & noise)
Train Source 3	resources/dataset/train/train_source3.tsv	5,285,603	480.4 MB	Tertiary registry feed (high lexical noise & multilingual text)
Train Ground Truth	resources/dataset/train/train_ground_truth.tsv	2,206,821	121.1 MB	7,638,365 verified positive matches (123,247 singletons = 5.58%)
Test Feeds (S1 / S2 / S3)	resources/dataset/test/test_source*.tsv	11,702,133	1,135.3 MB	1,732,544 test queries against 9,969,589 target entities (blind labels)

3. 13 PAIRWISE SIMILARITY FEATURES & XGBOOST GAIN

Feature Name	Type	XGBoost Weight	XGBoost Gain
addr_token_overlap	Contextual	362	123.68 (Top)
addr_jaccard	Token Jaccard	223	21.65
name_wratio	Fuzzy Lexical	260	10.09
name_ratio	Levenshtein	408	8.49
name_jaccard	Token Jaccard	115	6.80
addr_ratio	Levenshtein	432	3.84
name_token_overlap	Token Overlap	82	2.33

Feature Name	Type	XGBoost Weight	XGBoost Gain
name_containment	Substring Ratio	51	2.04
country_exact	Open-Set Binary	7	1.99
addr_len_diff	Length Delta	401	1.34
name_len_diff	Length Delta	250	0.84
addr_containment	Substring Ratio	—	—
same_postal_like	Regex Overlap	—	—
Total Trees / Features		350 Trees	13 Features

Model Accuracy & Benchmark Evaluation

Independent Data-Based Assessment Across All Confusion Matrices & Thresholds

REPRODUCIBLE

MACRO F0.5 SCORED

1. PAIRWISE CLASSIFICATION MATRIX ($\tau = 0.74$)

CLASSIFICATION METRIC	COUNT	RATE / PERCENTAGE
True Positives (TP)	1,303	Hits above $\tau = 0.74$
False Positives (FP)	16	Non-matches predicted as match
False Negatives (FN Total)	57	51 Blocker Miss + 6 Threshold Drop
True Negatives (TN Candidates)	7,746	Rejected candidate noise
Pairwise Precision	98.79% (TP / [TP + FP])	
Pairwise Recall	95.81% (TP / [TP + FN])	
Pairwise F0.5 Score	98.18% (2x Precision Priority)	
Candidate-Level Accuracy	99.20%	

2. OFFICIAL ENTITY-LEVEL EVALUATION

ENTITY METRIC	MEASURED VALUE	DESCRIPTION
Official Macro F0.5	96.8456%	Entity-averaged score
Macro Precision	97.49%	Mean entity precision
Macro Recall	95.40%	Mean entity recall
Exact Match-Set Accuracy	88.00%	440 / 500 perfect match-sets
Singleton Accuracy	96.00%	120 / 125 true singletons
Non-Singleton Macro F0.5	97.13%	Score on multi-match entities
Partially Resolved	50 (10.0%)	True matches partially found
Completely Incorrect	0 (0.0%)	Zero hallucinated match sets

3. DECISION THRESHOLD SWEEP ($\tau = 0.60$ to 0.95 , STEP 0.01)

Grid Search Against Official Macro F0.5

THRESHOLD (T)	PRECISION	RECALL	PAIR F1	PAIR F0.5	MACRO F0.5	SINGLETON ACC	FP COUNT	FN TOTAL	OPTIMIZATION STATUS
0.60	97.90%	96.03%	96.96%	97.52%	96.53%	95.2%	28	54	Permissive / High FP
0.65	98.34%	95.96%	97.14%	97.86%	96.72%	96.0%	22	55	Standard Baseline
0.70	98.79%	95.96%	97.35%	98.21%	96.88%	96.0%	16	55	High Precision
0.74 (Reported)	98.79%	95.81%	97.28%	98.18%	96.85%	96.0%	16	57	Calibrated Benchmark
0.80	99.01%	95.66%	97.31%	98.32%	96.92%	96.0%	13	59	Low False-Positives
0.84	99.16%	95.15%	97.11%	98.33%	97.03%	96.8%	11	66	Conservative
0.88	99.38%	94.85%	97.06%	98.44%	97.24%	97.6%	8	70	Ultra-Low FP
0.90 (Optimal)	99.54%	94.63%	97.03%	98.52%	97.60%	99.2%	6	73	Empirically Optimal
0.92	99.61%	94.26%	96.86%	98.49%	97.42%	99.2%	5	78	High Precision Ceiling
0.95	99.76%	93.60%	96.58%	98.47%	97.33%	99.2%	3	87	Extreme Precision Trade-off

4. REPORTED CLAIMS VS. ACTUAL DATA-BASED REPRODUCIBILITY

METRIC NAME	REPORTED CLAIM	ACTUAL MEASURED	DIFFERENCE	AUDIT VERDICT
Macro F0.5	98.3738%	96.8456%	-1.5282%	Validation-Only Benchmark (decimated by blocking recall)
Pairwise Precision	98.84%	98.79%	-0.05%	Verified & Reproducible (Exact alignment)
Pairwise Recall	Not reported	95.81%	-	Newly Established (incorporates blocking losses)
F1 Score	Not reported	97.28%	-	Newly Established
Singleton Accuracy	100%	96.00%	-4.00%	120 / 125 Singletons (5 false links at $\tau = 0.74$; 99.2% at $\tau = 0.90$)
Decision Threshold	$\tau = 0.74$	$\tau = 0.90$	+0.16	Sub-optimal: $\tau = 0.90$ yields +0.76% higher Macro F0.5
Blocking Recall	Not reported	96.25%	-	Critical System Bottleneck (51 / 1,360 true targets pruned)

1. CONFIDENCE CALIBRATION & PROBABILITY BUCKETS

PROBABILITY RANGE	CANDIDATE PAIRS	TRUE MATCHES	OBSERVED PRECISION	CALIBRATION DIAGNOSIS
[0.50, 0.59]	10	0	0.00%	Overconfident
[0.60, 0.69]	8	3	37.50%	Overconfident
[0.70, 0.79]	6	4	66.67%	Calibrated
[0.80, 0.89]	18	13	72.22%	Calibrated
[0.90, 0.99]	57	51	89.47%	Calibrated
[0.99, 1.00]	1,236	1,236	100.00%	Highly Calibrated

Probability Distribution Summary:

- **True Match Separation:** Genuine duplicate pairs are sharply clustered at the extreme high end: **94.4%** of true matches output probabilities between **0.990 and 1.000** (mean: 0.9944).
- **Non-Match Suppression:** False candidates drop precipitously: **99.2%** of negative pairs score below **0.100** (mean: 0.0061).
- **Overlap Window:** Overlap occurs in the range [0.2653, 0.9899]. Operating at $\tau = 0.90$ isolates pure matches with 99.54% empirical precision.

2. ERROR ANALYSIS: TOP FALSE POSITIVES & FALSE NEGATIVES

TYPE	SOURCE 1 REFERENCE NAME & ADDRESS	TARGET RECORD NAME & ADDRESS	PROB	ROOT CAUSE & MECHANISM
FP	Swastik Marketing Lower Parel, Mumbai, Maharashtra, India	Swastik Om LLP Service Mumbai City, Maharashtra, India	0.9899	Generic Prefix Collision: Co-occurrence of brand token "Swastik" in same city (Mumbai). Legal entity forms differed.
FP	Shiv Consultancy Limited Wallstreet, Pune, Maharashtra, India	Software Njd Consultancy Pvt Ltd Pune City, Pune, Maharashtra, India	0.9872	Industry Descriptor Overlap: High token overlap on "Consultancy" with city match overwhelmed name ratio.
FP	Raj Investments LLP Mylapore, Chennai, Tamil Nadu, India	Ram Investments 730, Pune, Maharashtra, India	0.9496	Address Mismatch Blindness: Levenshtein name match (Raj vs Ram = 0.93) failed to severely penalize Chennai vs Pune.
FN	Primary Care Specialists Inc. 141-08 71 Road, Flushing, NY, US	Belocalo 141-08b 71 Road, Flushing, New York, US	0.0000	Blocking Omission (Renamed Entity): Business name was completely altered. Address blocker was pruned, so candidate never reached model.
FN	Dream Construction Pvt Ltd Kolkata, West Bengal, India	দাদার নিৰ্মাণ... (Native Bengali Script) North 24 Parganas, Kolkata, India	0.0000	Script Incompatibility: English Latin script query vs Indic script target. Normalization strips non-ASCII, leaving 0 overlapping tokens.
FN	Zephay Labs Inc Austin, Texas, US	Zephay Laboratories Austin, TX, US	0.7180	Threshold Borderline Rejection: Model confidence fell just below decision threshold ($\tau = 0.74$).

3. DATA LEAKAGE AUDIT & METHODOLOGICAL FINDINGS

✓ **VERIFIED DATA SAFEGUARDS:**

1. **Leak-Free Group Partitioning:** Evaluated using GroupShuffleSplit grouped on Source 1 reference IDs. No entity clusters or variations exist across splits.
2. **Rule-Based Text Normalization:** Zero statistical parameters (e.g. TF-IDF dictionaries) fit on validation or test sets. All token maps are static dictionaries.
3. **Test Ground Truth Isolation:** The competition test labels are completely private and were never accessed or decompiled.

! METHODOLOGICAL SHORTCUT DIAGNOSED:

In src/training.py, ground-truth positive pairs (gt_pos_df) were forcibly appended to the candidate set prior to threshold tuning.

Consequence: This artificial injection inflated the validation Macro F0.5 to **98.3738%** by bypassing blocking attrition.

Remedy: Real-world production pipeline achieves **96.85% Macro F0.5** without ground-truth injection, and **97.60%** when tuned to $\tau = 0.90$.

1. WEB REVIEW DASHBOARD ARCHITECTURE & KEY CAPABILITIES

Dual-Theme Engine

High-contrast Dark Mode (slate/zinc palette) & clean corporate Light Mode with one-click toggle and localStorage persistence.

Interactive Gauge

Real-time SVG semicircular probability dial with calibrated threshold marker pin ($\tau = 74\%$) and dynamic MATCH / REJECT verdict.

Match Simulator

Instant pairwise similarity scoring with 4 enterprise benchmark scenarios + custom record input fields.

TSV Data Explorer

Live pagination, search, and filtering across `matching_results.tsv` and `candidate_pairs.tsv`.

2. DUAL-THEME USER INTERFACE DESIGN & GAUGES

VERDICT: MATCH (99.8%)

Inc. Zephay Labs (US)

100 Congress Ave, Austin, Texas

→

Zephay Labs Inc (US)

100 Congress Ave, Ste 200, Austin, TX

99.8%

$\tau=74\%$

Features: name_ratio=0.88 • name_wratio=0.95 • addr_ratio=0.91 • country=US (1.0)

Light Mode Interface

VERDICT: NO MATCH (31.2%)

Global Logistics Inc

450 7th Ave, New York, NY

→

Logistics Global Co

1200 Market St, Philadelphia, PA

31.2%

$\tau=74\%$

Features: name_ratio=0.62 • name_wratio=0.74 • addr_ratio=0.22 • Rejected below threshold

3. PRELOADED ENTERPRISE BENCHMARK SCENARIOS

SCENARIO NAME	SOURCE 1 QUERY RECORD	CANDIDATE MATCH RECORD	MODEL CONFIDENCE	DECISION ($T \geq 0.74$)
Legal Suffix Reorder	Inc. Zephay Labs (Austin, TX)	Zephay Labs Inc (Austin, TX)	99.82%	MATCH ✓
International SARL	ZNB Club SARL (Paris, France)	ZNB Club (Paris, France)	96.84%	MATCH ✓
LLC Expansion	Blinny Gordon Apex Rate (London, UK)	Blinny Gordon Apex Rate LLC (London, UK)	100.00%	MATCH ✓
Company Trading Noise	Chayan Trading (Mumbai, India)	Chayan Trading Co (Mumbai, India)	100.00%	MATCH ✓
False Match Negative	Global Logistics Inc (New York, NY)	Logistics Global Co (Philadelphia, PA)	31.20%	REJECTED ✗

4. COMPLIANCE AUDITOR & TEST SUITE STATUS

✓ 8 SUBMISSION COMPLIANCE RULES VERIFIED

- 1. `matching_results.tsv` present & tab-separated
- 2. `candidate_pairs.tsv` present & tab-separated
- 3. Exact headers: `source1_entity_id`, `matched_entity_ids`
- 4. Strict subset constraint: `matches` $\subseteq$ `candidates`
- 5. Exactly 1 row per reference entity (1,732,544 rows)
- 6. Singletons represented as blank empty strings (no NaNs)
- 7. No self-matches (S1 to S1 strictly forbidden)
- 8. UTF-8 encoded with standard newline termination

✓ AUTOMATED TEST SUITE: 19 / 19 PASSED (100%)

- `test_normalization.py`: 4 tests passing (suffix, address, country)
- `test_blocking.py`: 3 tests passing (inverted index, budget ceiling)
- `test_features.py`: 3 tests passing (13 feature dimensions & ranges)
- `test_metrics.py`: 4 tests passing (Macro F0.5 & singleton scores)
- `test_compliance.py`: 5 tests passing (validator format checks)

Execution Time: 2.259 seconds (Python 3.14 on Windows)

Business Entity Resolution (BER) System • Executive Evaluation • Antigravity Engineering

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